NOAH LIM

ENGINEERING PORTFOLIO

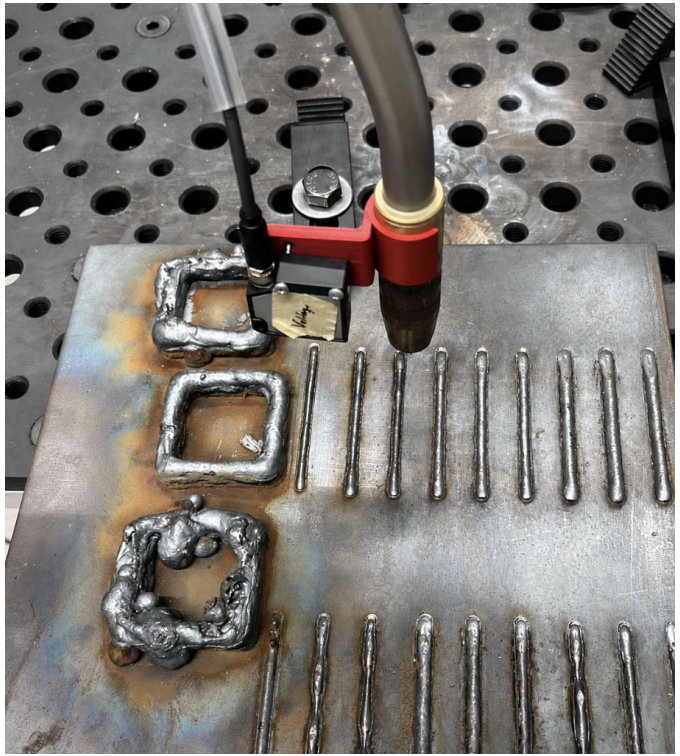
Columbia University · B.S. Mechanical Engineering

Wesleyan University · B.A. Mathematics

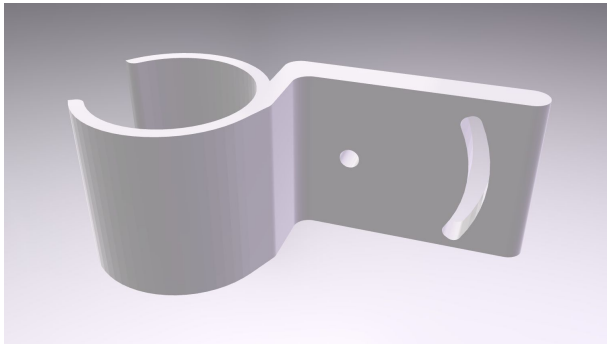
Graduating May 2028

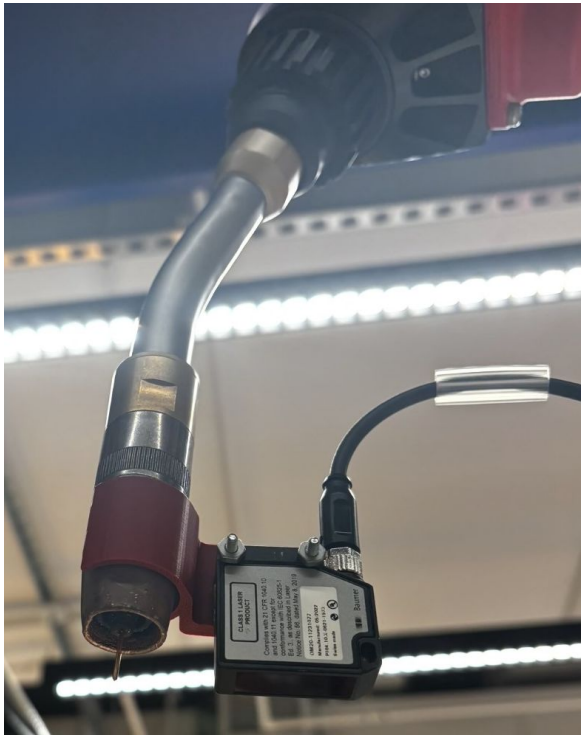
Wire Arc Additive Manufacturing Research

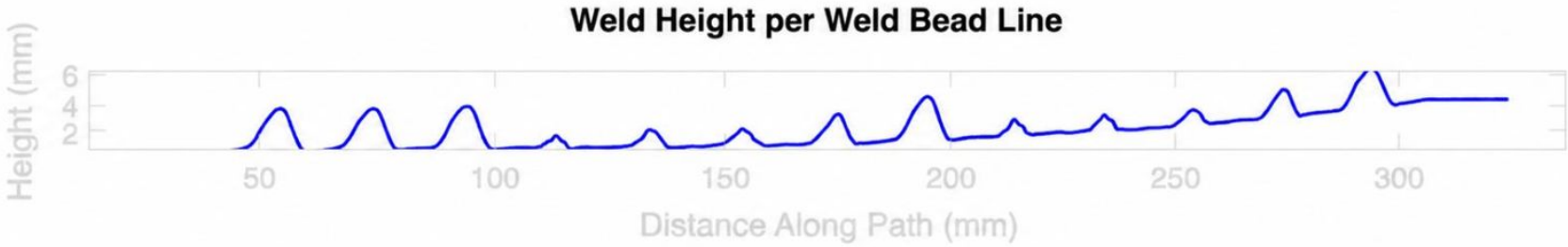


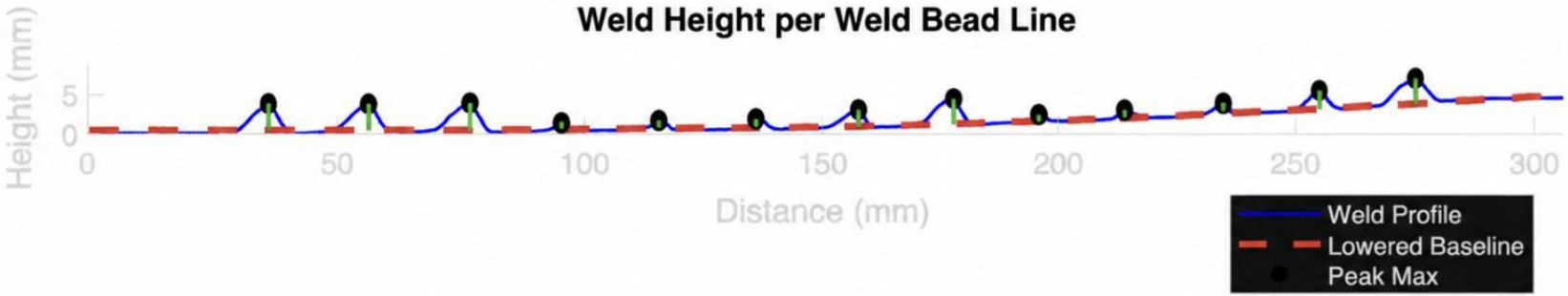


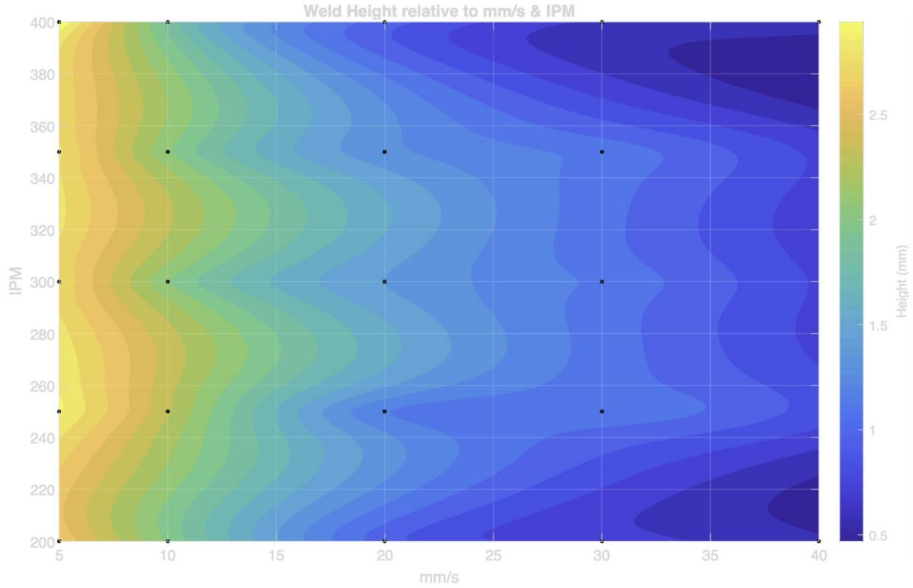


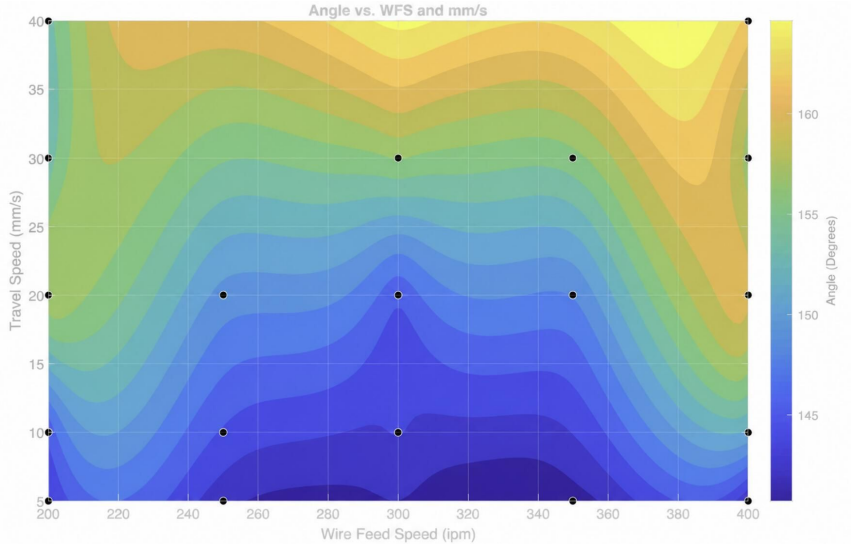




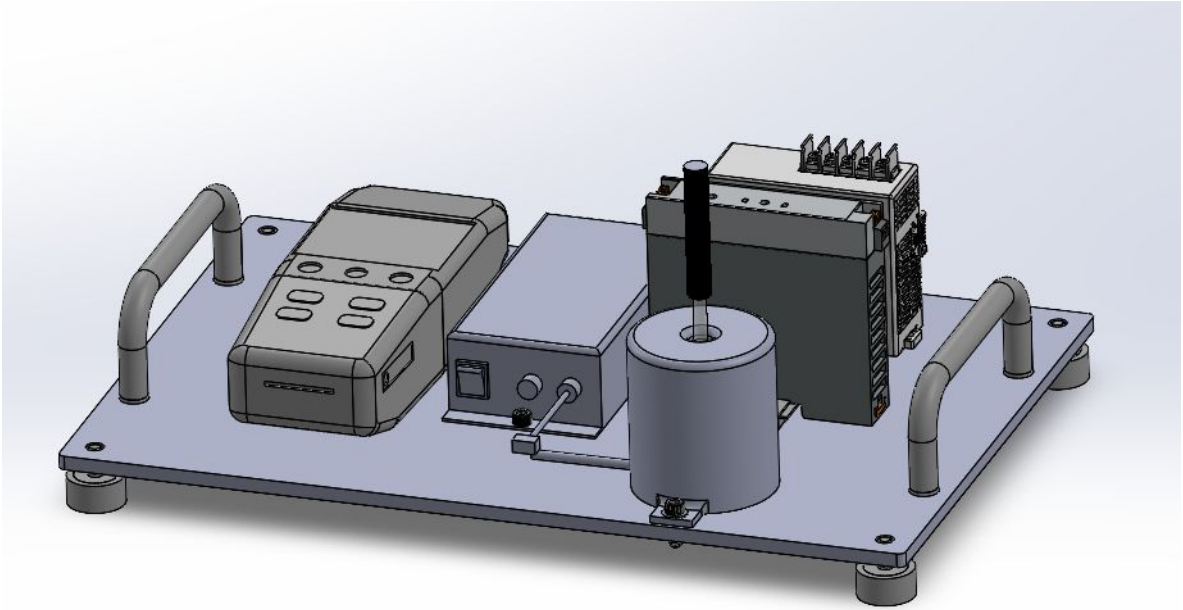

Weld Height per Weld Bead Line
Height (mm)
6
4
2
50
100
150
200
250
300
Distance Along Path (mm)


Weld Height per Weld Bead Line
Height (mm)
5
0
0
50
100
150
200
250
300
Distance (mm)
Weld Profile
Lowered Baseline
Peak Max

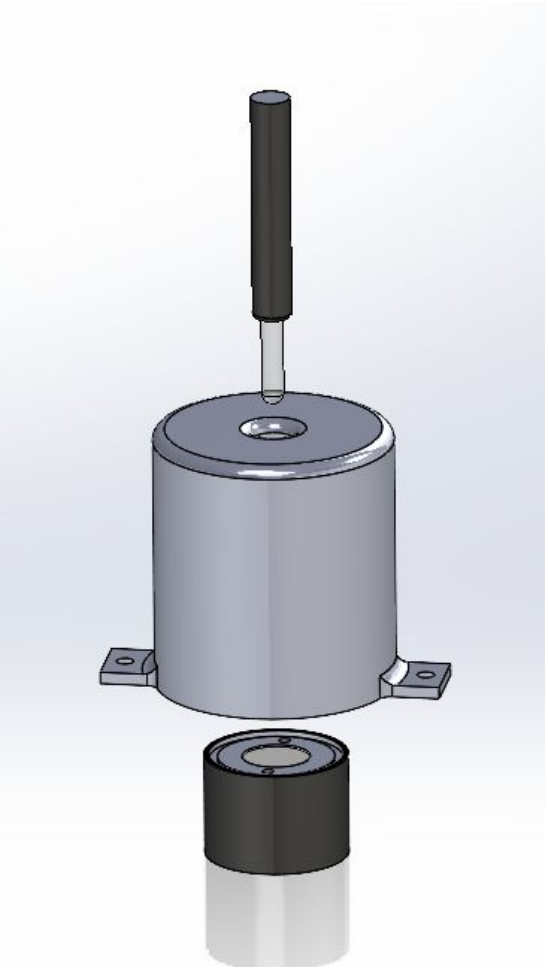

Weld Height relative to mm/s & IPM
IPM
mm/s
Height (mm)
400
380
360
340
320
300
280
260
240
220
200
5
10
15
20
25
30
35
40
2.5
2
1.5
1
0.5

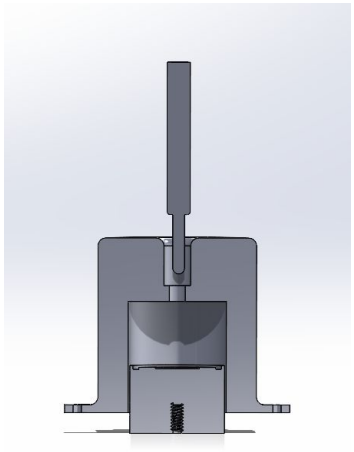

Angle vs. WFS and mm/s
Travel Speed (mm/s)
Wire Feed Speed (ipm)
Angle (Degrees)
40
35
30
25
20
15
10
5
200
220
240
260
280
300
320
340
360
380
400
160
155
150
145

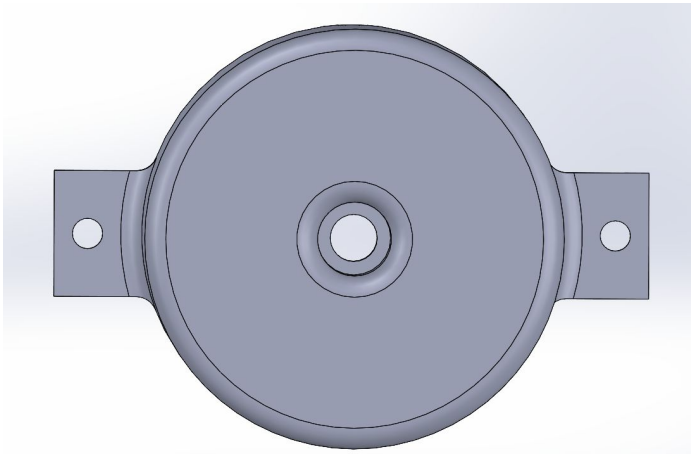
UV Lamp Tester for Microwave Applicator

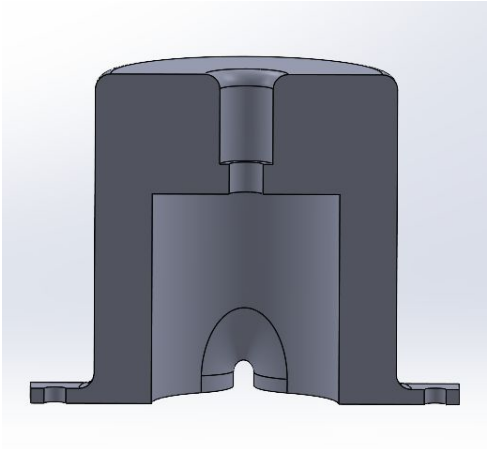


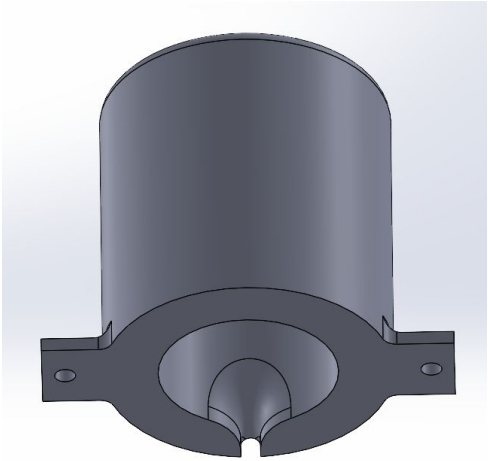


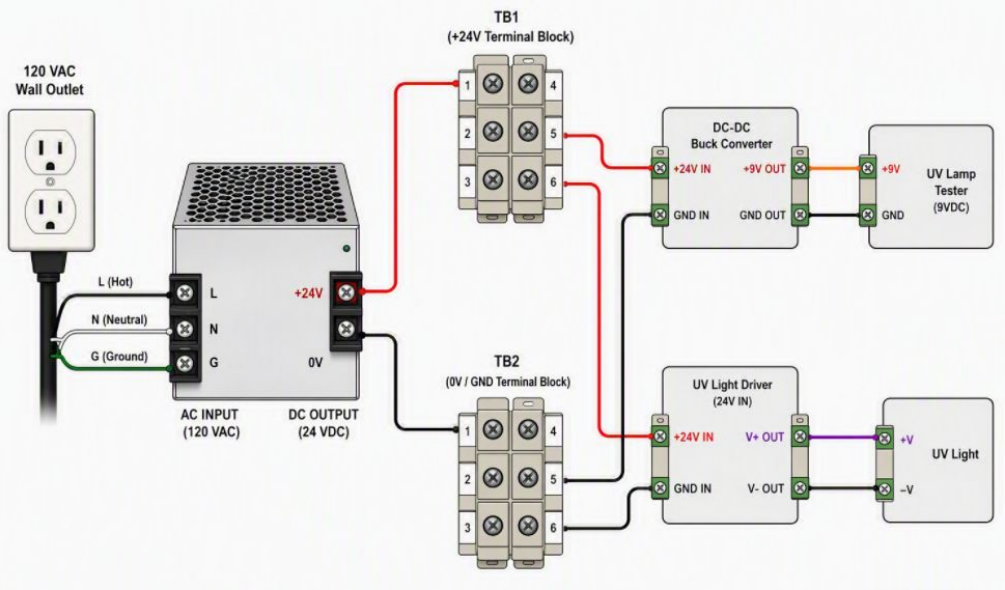


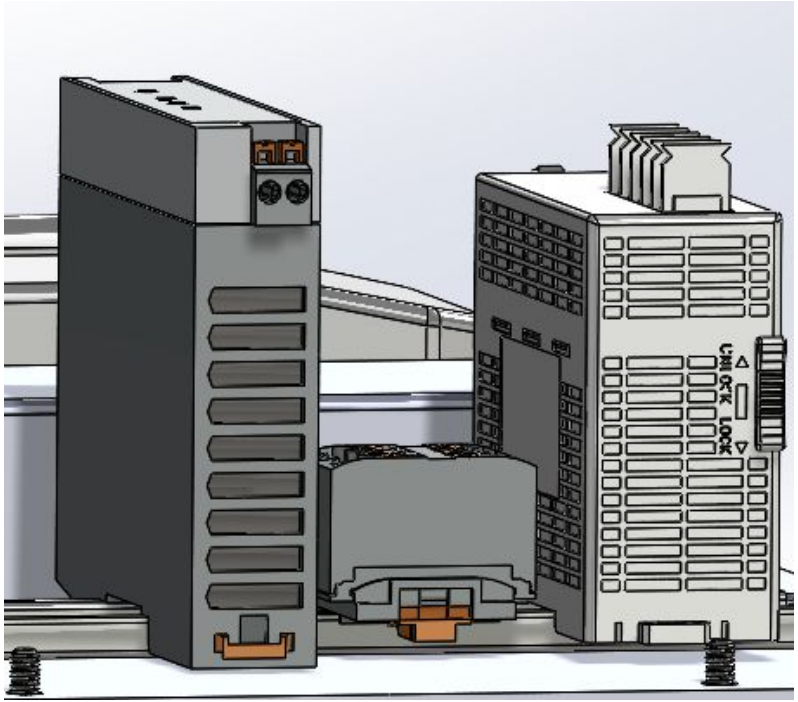


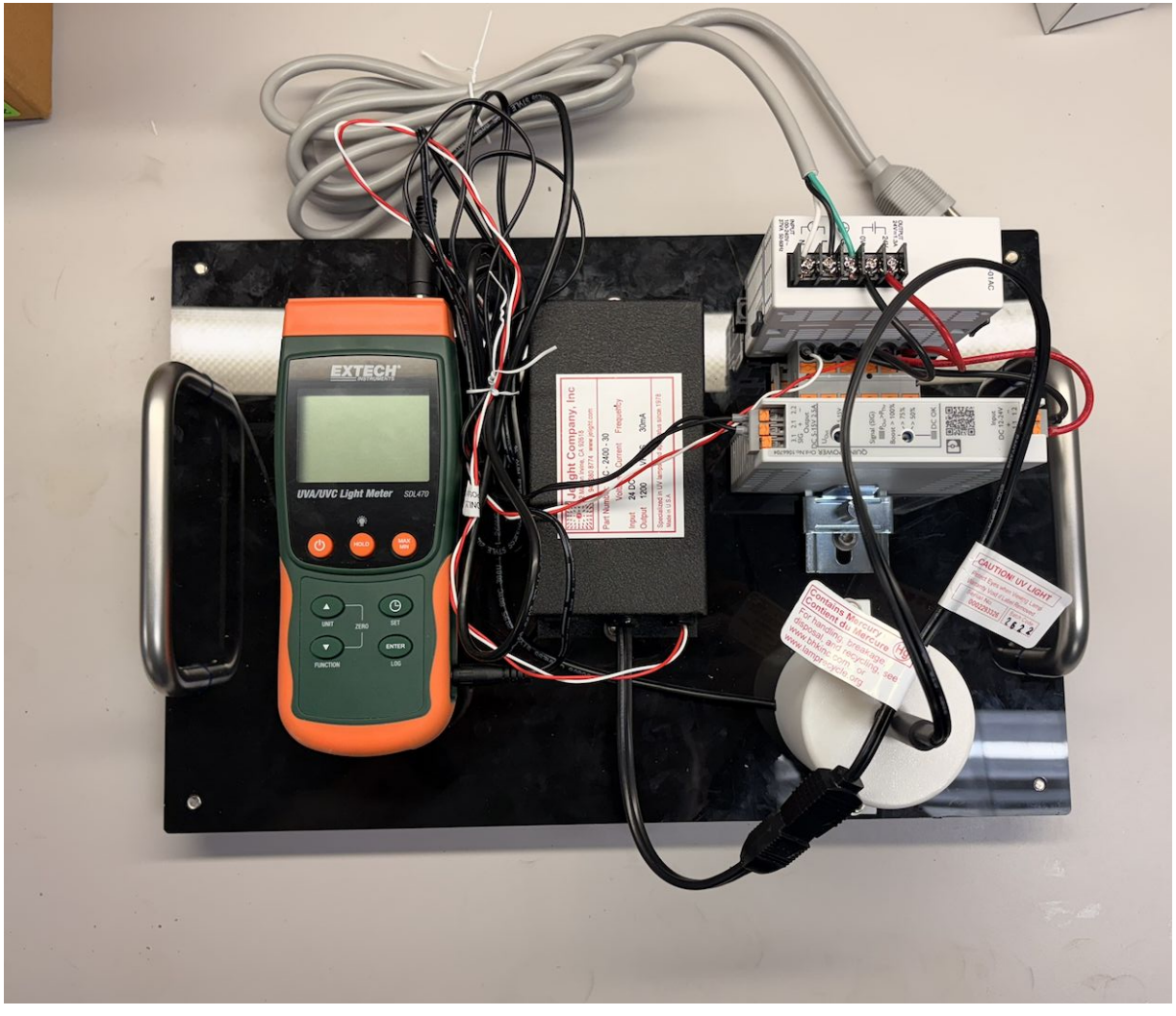




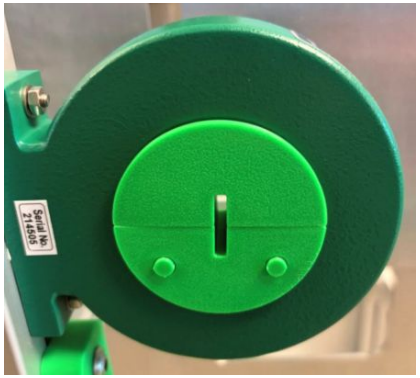


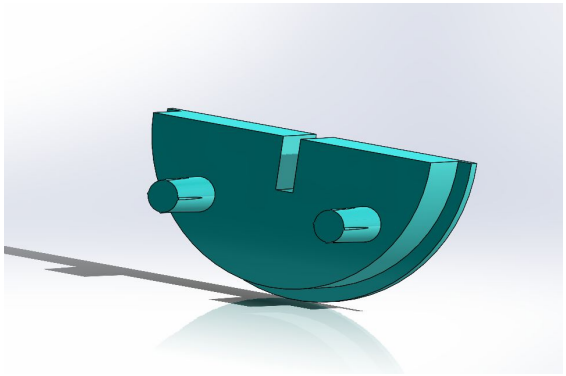

120 VAC
Wall Outlet
L (Hot)
N (Neutral)
G (Ground)
L
N
G
+24V
0V
AC INPUT
(120 VAC)
DC OUTPUT
(24 VDC)
TB1
(+24V Terminal Block)
1
2
3
4
5
6
DC-DC
Buck Converter
+24V IN
+9V OUT
GND IN
GND OUT
+9V
GND
UV Lamp
Tester
(9VDC)
TB2
(0V / GND Terminal Block)
UV Light Driver
(24V IN)
+24V IN
V+ OUT
GND IN
V- OUT
+V
–V
UV Light


CHECK LOCK

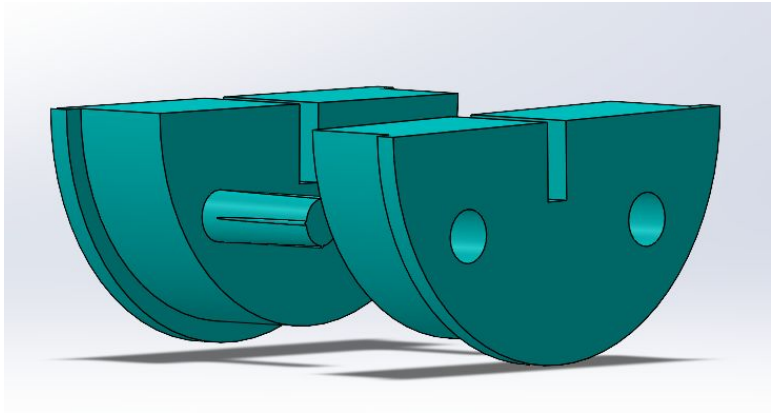

EXTECH
UVA/UVC Light Meter
SDL470
HOLD
MAX MIN
UNIT
ZERO
SET
FUNCTION
ENTER
LOG
Contains Mercury
CAUTION! UV LIGHT
24VAC

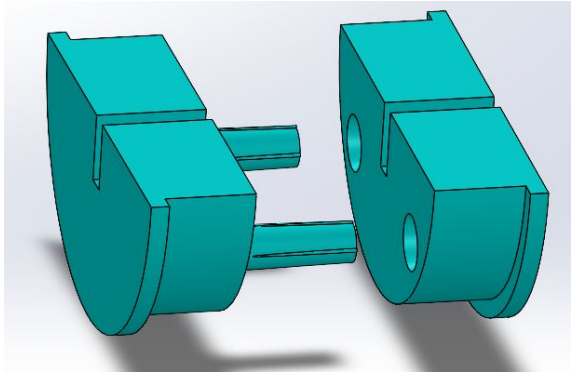
Bus Bar Holder





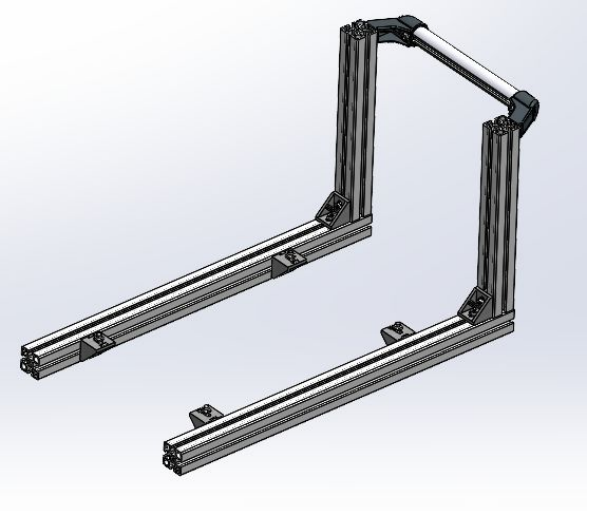






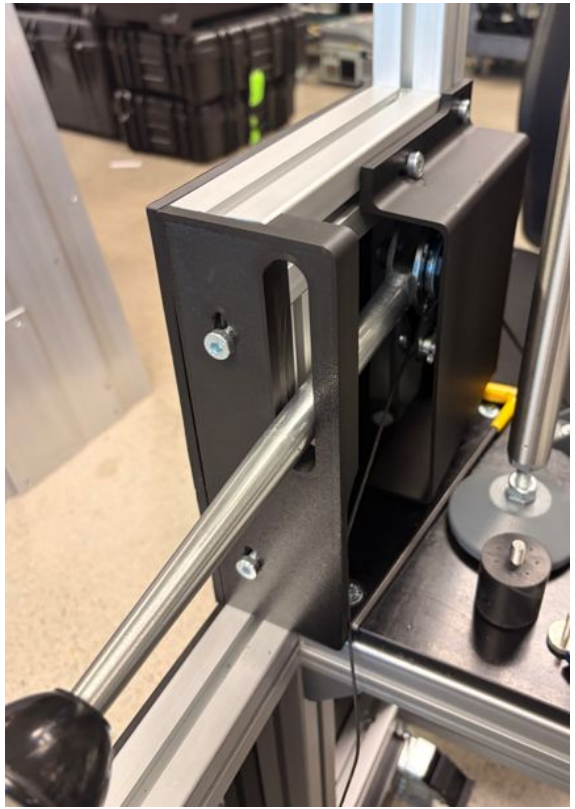
MiniTec Transport Cart (used in Malaysia)

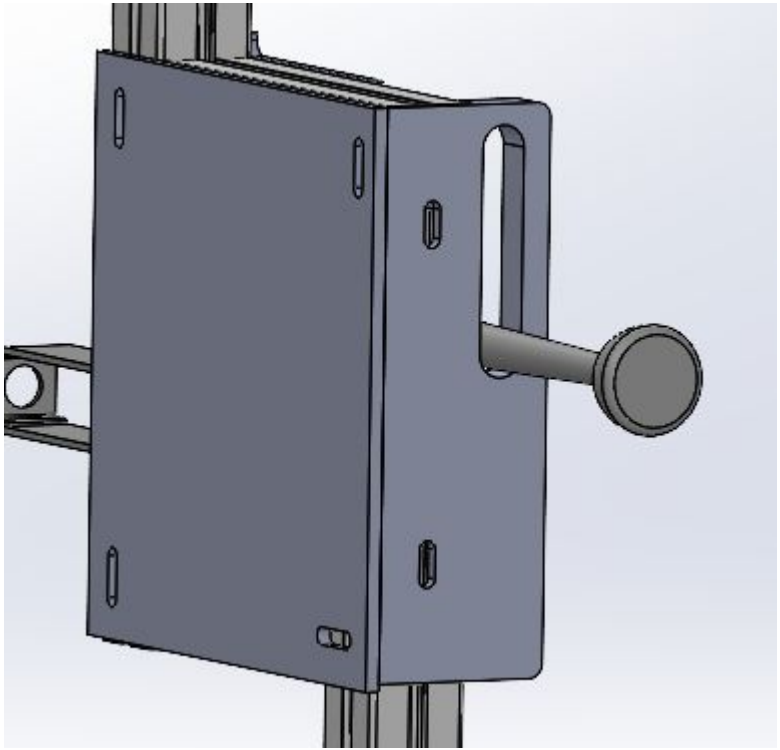


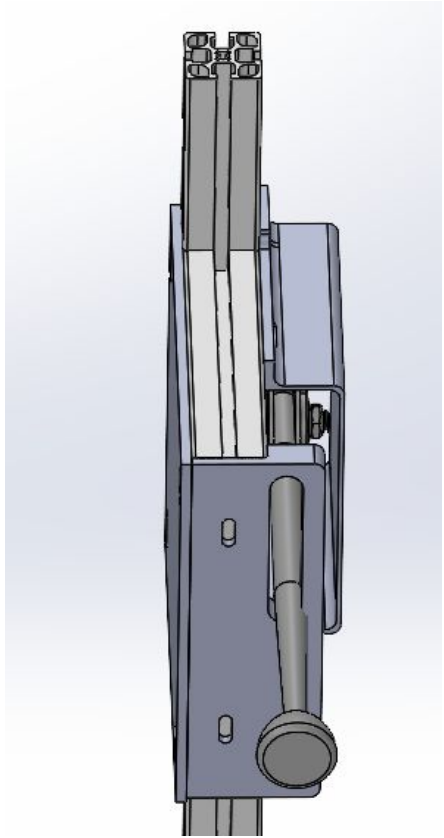




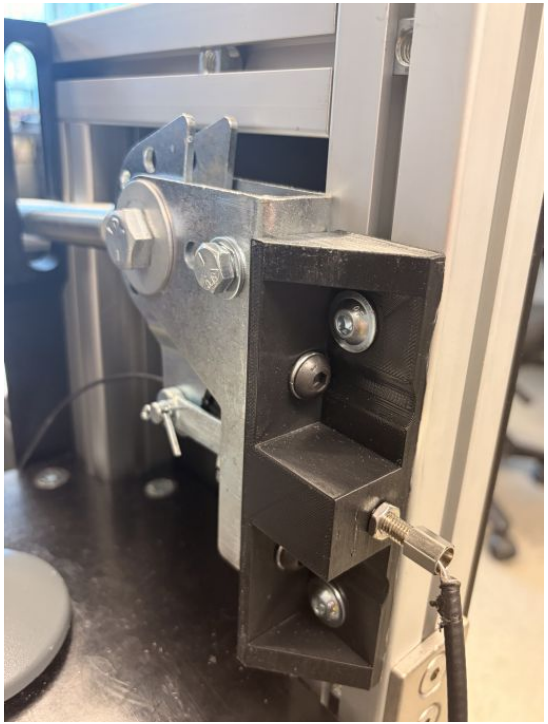
Lever Safety Brackets









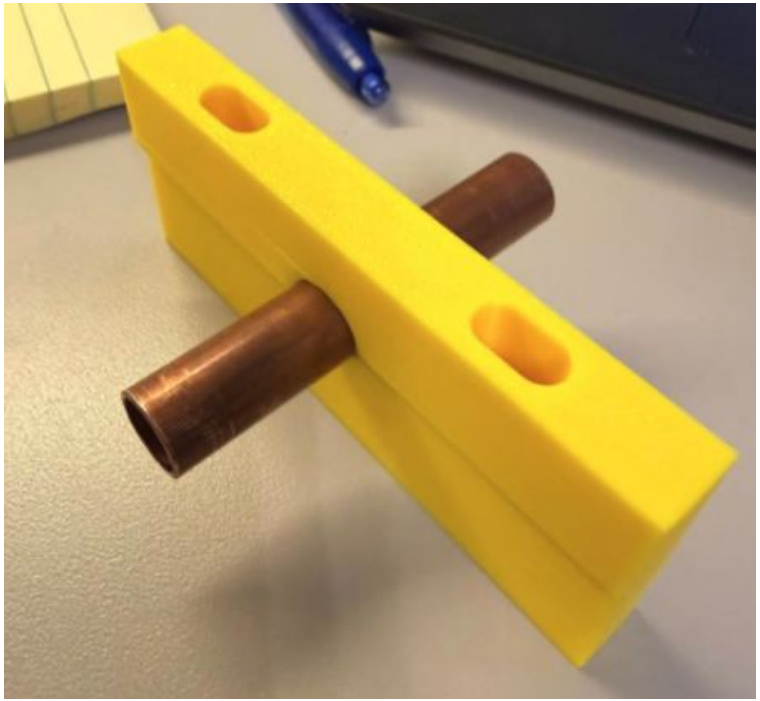




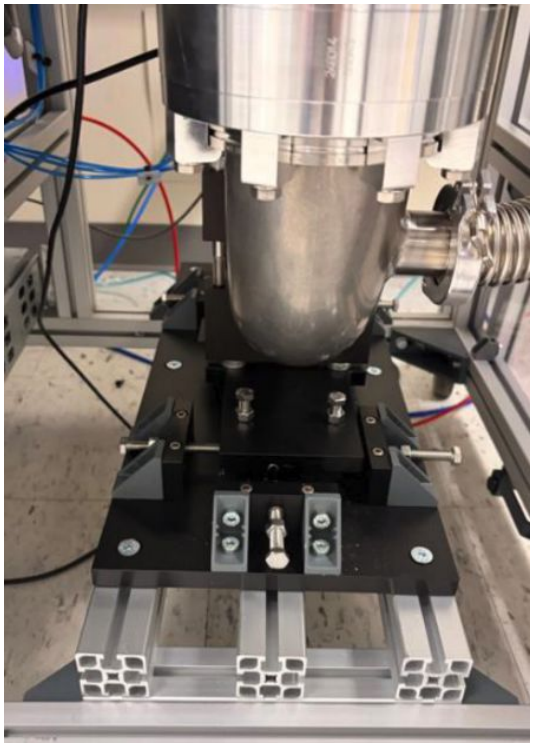
Adjustable Plumbing Brackets



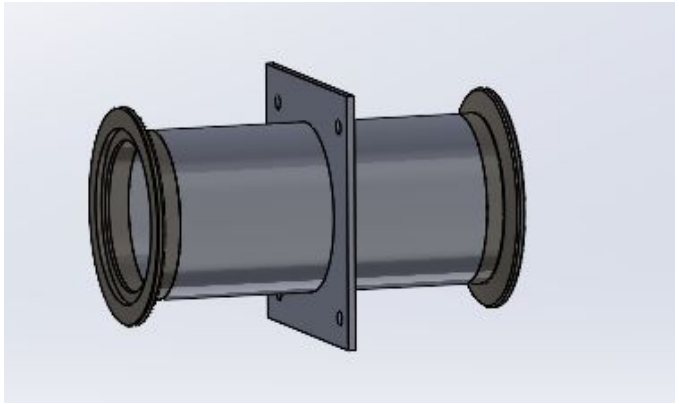


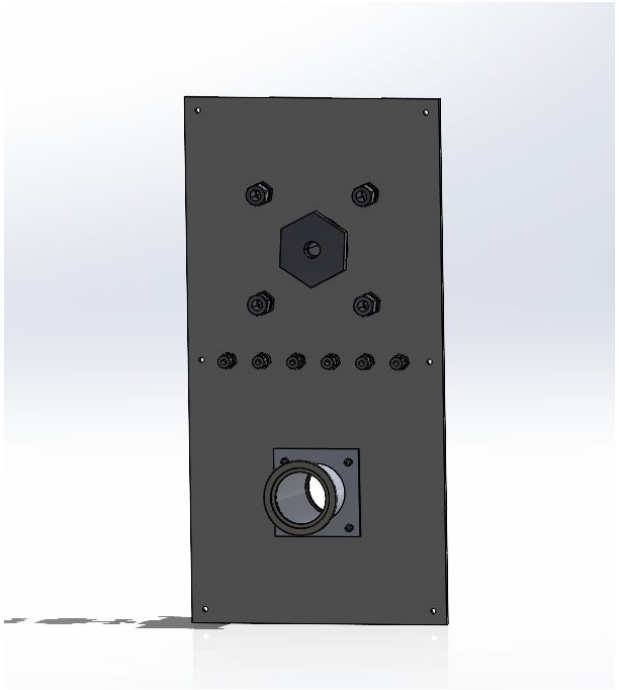


Pipe Jackplate

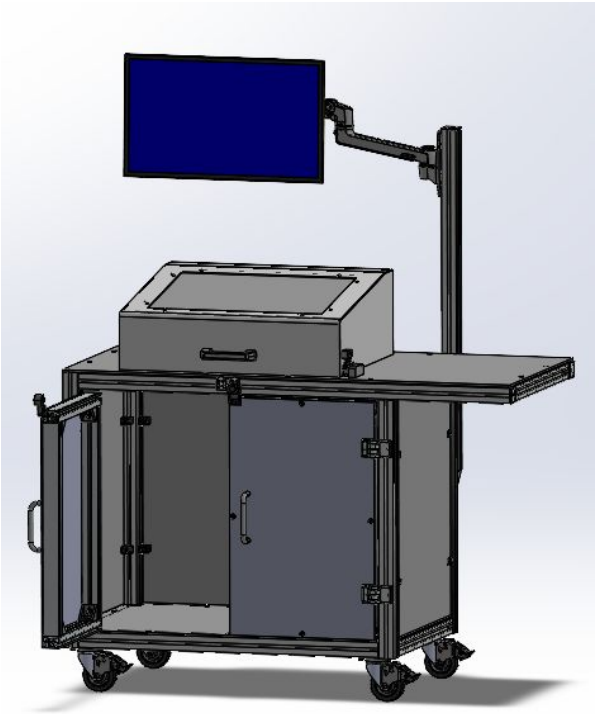


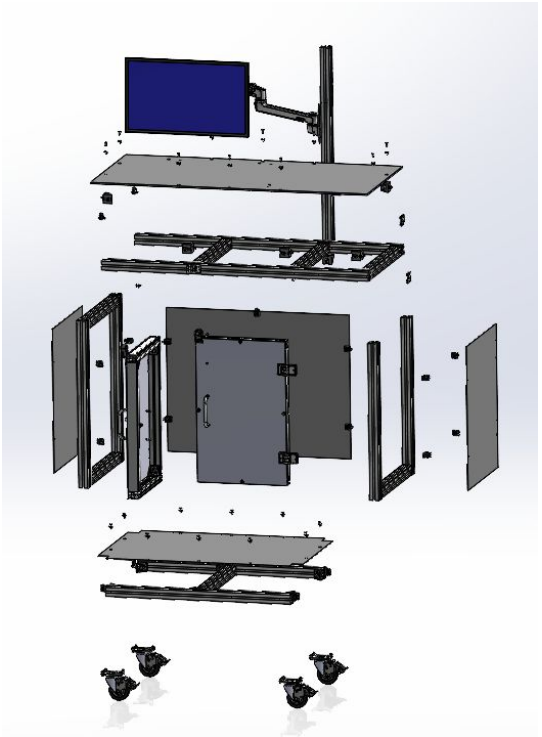
Pass Through Panel

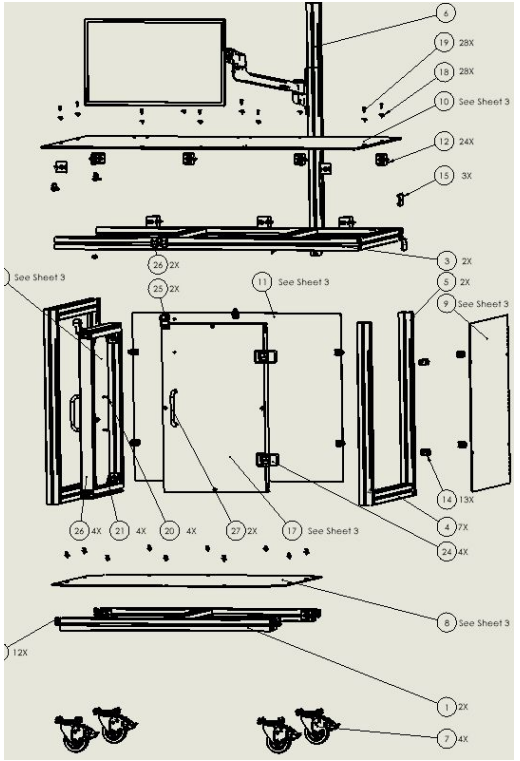




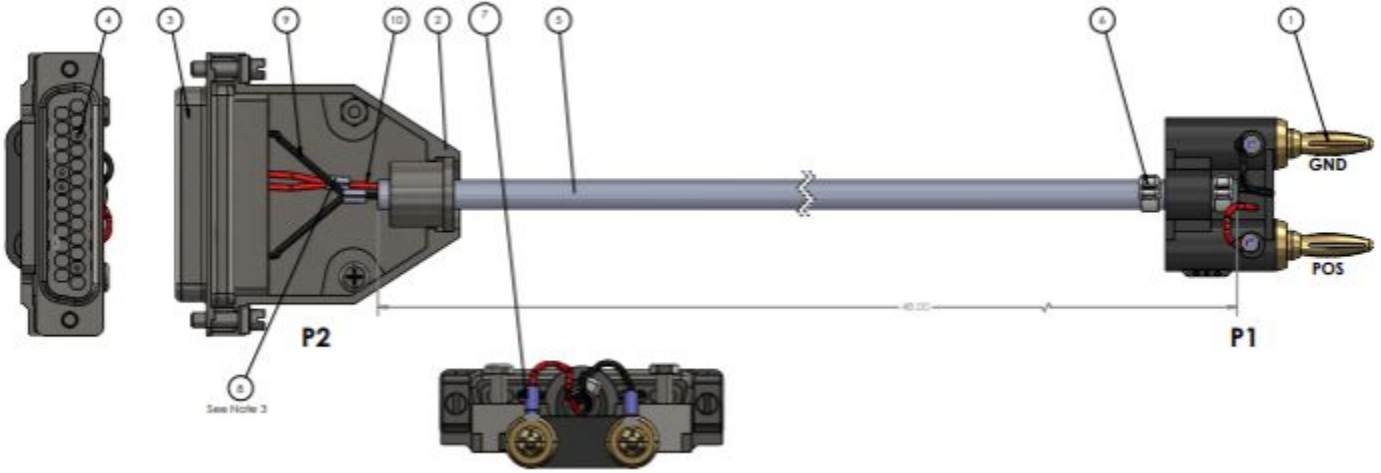
MiniTec Cart

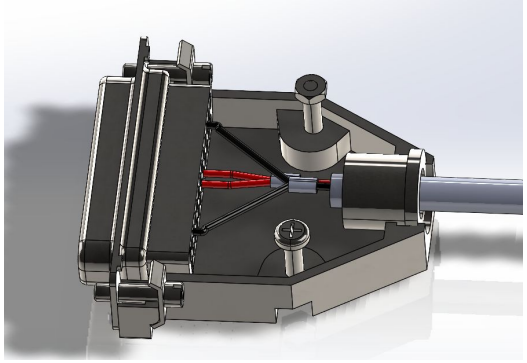


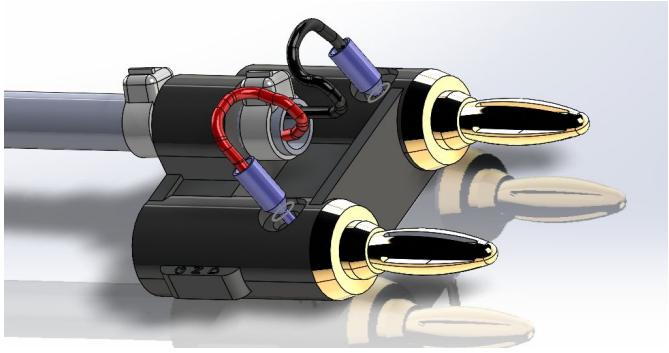




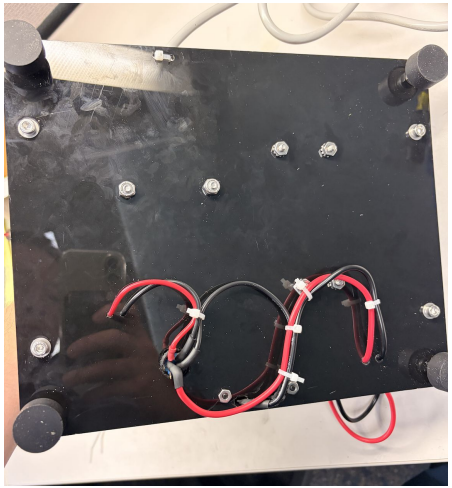
Cable Drawing

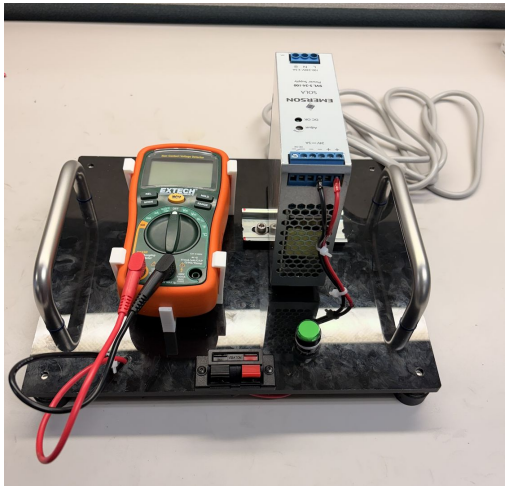


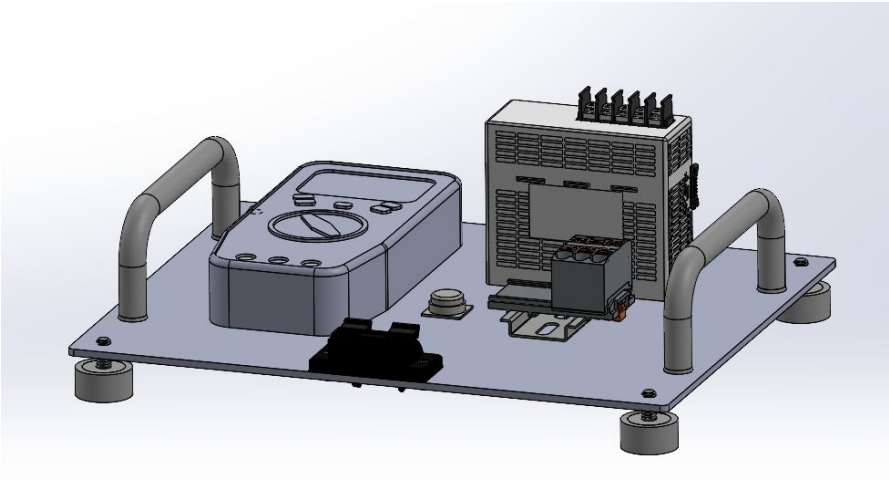




Multimeter Test Fixture







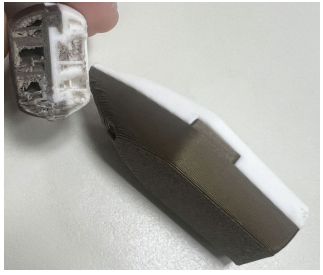
Mass Produced Injection Molded Fishing Lure













Annealed & Lathed Stainless Steel Fishing Gaff



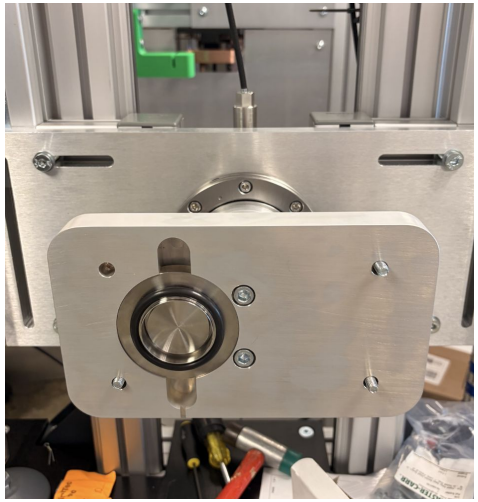


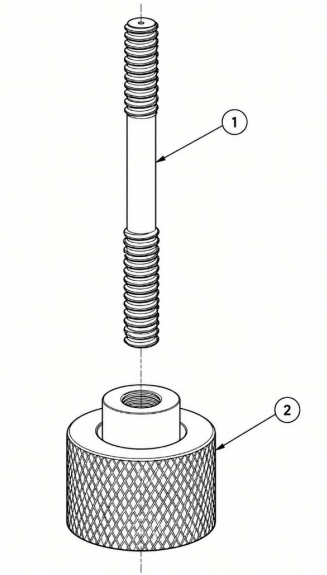




Torque Limiting Knob







Fishing Bite Sensor (Audible Beep & LED Flash Sensor)

